

CHONGJIN, Korea, Democratic People's Republic of

WMO: 470080

Lat: 41.783N

Lon: 129.817E

Elev: 141

StdP: 14.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.8	12.6	-15.1	2.4	16.7	-10.0	3.2	17.6	22.0	19.1	19.3	17.6	6.5	320	0.400

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.9	81.9	73.2	79.6	72.2	77.5	71.3	75.8	79.5	74.4	77.9	72.9	76.2	3.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.6	130.0	78.4	73.1	123.7	76.9	71.7	117.8	75.6	39.5	80.1	38.1	77.8	36.8	76.2	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.7	13.6	10.9	DB	4.6	87.6	4.0	2.5	1.7	89.4	-0.6	90.9	-2.9	92.3	-5.8	94.1	
			WB	2.5	78.1	3.4	2.6	0.1	80.0	-1.9	81.5	-3.7	83.0	-6.2	84.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	23.1	26.4	35.2	46.2	54.2	62.0	69.5	72.1	64.3	53.3	38.8	27.8
	DBStd	17.55	5.55	5.83	6.09	5.93	5.80	5.23	5.07	3.86	4.34	5.60	7.30	6.23
	HDD50	3183	833	661	459	145	24	0	0	0	32	340	689	
	HDD65	6708	1298	1081	923	565	338	120	16	62	363	787	1154	
	CDD50	2401	0	0	1	29	154	359	605	686	430	134	3	0
	CDD65	452	0	0	0	0	3	29	156	222	42	0	0	0
	CDH74	1542	0	0	0	0	15	95	532	818	82	0	0	0
	CDH80	171	0	0	0	0	2	10	65	88	6	0	0	0
Wind	WSAvg	3.2	3.7	3.5	3.5	3.0	2.8	2.4	2.2	2.6	2.9	3.6	3.8	3.9
Precipitation	PrecAvg	25.3	0.4	0.4	0.6	1.7	2.7	3.4	5.6	5.0	3.5	1.4	1.0	0.6
	PrecMax	40.5	1.5	1.7	1.9	8.7	9.3	8.6	11.3	18.0	9.4	3.6	3.6	2.8
	PrecMin	18.0	0.0	0.0	0.0	0.0	0.0	0.3	0.9	0.4	0.4	0.1	0.0	0.0
	PrecStd	5.2	0.4	0.4	0.5	1.8	2.0	2.0	3.0	3.4	2.6	1.1	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.4	44.9	57.9	69.4	75.7	81.3	85.4	85.6	80.2	69.7	59.4	46.0
		MCWB	33.8	37.4	46.1	53.0	59.4	67.4	75.2	75.4	68.0	59.4	50.9	40.8
	2%	DB	35.9	40.6	51.8	63.2	70.4	76.5	81.5	82.3	76.3	67.1	55.9	42.3
		MCWB	31.0	34.9	42.2	50.8	57.6	65.2	73.4	74.5	67.2	59.2	49.2	37.4
	5%	DB	33.5	37.8	47.8	59.0	66.9	73.0	79.1	80.2	74.0	64.9	52.8	39.5
		MCWB	29.5	33.1	39.9	48.4	55.9	64.1	72.5	73.3	66.3	57.6	46.7	34.7
	10%	DB	31.4	35.4	44.7	55.6	63.5	69.7	76.9	78.2	72.0	62.7	50.0	36.9
		MCWB	27.8	31.3	38.0	46.6	54.4	62.9	72.1	72.5	65.1	56.0	44.4	32.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	35.1	38.5	47.2	55.3	62.0	70.2	77.2	78.3	72.6	64.3	52.9	41.7
		MCDB	38.3	42.8	57.0	65.2	71.0	77.5	81.8	82.1	76.7	67.4	57.3	44.8
	2%	WB	31.9	35.7	43.1	52.0	59.2	67.6	75.4	76.5	69.9	61.1	50.2	38.2
		MCDB	34.6	39.5	50.2	61.2	67.9	73.0	79.3	80.2	74.1	64.9	54.8	41.7
	5%	WB	30.0	33.7	40.9	49.6	57.3	65.7	74.0	75.1	68.1	59.3	47.7	35.5
		MCDB	32.9	37.3	46.7	57.6	64.3	70.4	77.6	78.7	72.1	63.4	52.0	38.9
	10%	WB	28.3	31.6	38.6	47.6	55.6	64.0	72.6	73.8	66.6	57.3	45.1	33.0
		MCDB	31.1	35.1	43.8	54.4	61.7	68.6	76.1	77.0	70.4	61.7	49.3	36.2
Mean Daily Temperature Range	5% DB	MDBR	11.6	11.9	11.8	12.7	11.6	9.0	7.8	8.9	12.8	14.3	12.5	11.7
		MCDBR	13.6	14.0	16.4	18.2	18.9	15.0	12.0	11.5	14.2	15.6	16.4	14.3
		MCWBR	11.6	11.5	10.8	10.3	9.8	7.7	6.6	6.8	8.7	11.5	13.2	12.0
	5% WB	MCDBR	13.0	13.6	15.3	17.0	16.0	11.7	10.0	9.6	11.7	14.2	15.7	13.9
		MCWBR	11.6	11.4	11.3	10.7	9.3	7.0	6.2	6.5	8.2	11.4	13.1	12.5
Clear-Sky Solar Irradiance	taub	0.337	0.390	0.460	0.540	0.578	0.583	0.540	0.493	0.438	0.434	0.390	0.347	
	taud	2.176	2.048	1.909	1.780	1.777	1.851	2.038	2.168	2.234	2.101	2.122	2.155	
	Ebn at Noon	247	250	245	235	230	229	238	246	251	232	224	230	
	Edh at Noon	32	42	55	67	69	64	53	45	39	40	34	30	
All-Sky Solar Radiation	RadAvg	709	1014	1363	1546	1592	1527	1396	1413	1334	1063	708	604	
	RadStd	45	66	88	109	161	200	174	109	120	75	90	43	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	-4.14	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	-1.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page